

Unit-1

Random Variables and Probability Distributions

Random variable

- A **random variable** is a function that associates a real number with each element in the sample space.
- **Example:**

Two balls are drawn in succession without replacement from an urn containing 4 red balls and 3 black balls. The possible outcomes and the values y of the random variable Y , where Y is the number of red balls, are

Sample Space	y
RR	2
RB	1
BR	1
BB	0



Examples of random variable

- Example:

Statisticians use **sampling plans** to either accept or reject batches or lots of material. Suppose one of these sampling plans involves sampling independently 10 items from a lot of 100 items in which 12 are defective.

Let X be the random variable defined as the number of items found defective in the sample of 10. In this case, the random variable takes on the values $0, 1, 2, \dots, 9, 10$. 

- Example:

Let X be the random variable defined by the waiting time, in hours, between successive speeders spotted by a radar unit. The random variable X takes on all values x for which $x \geq 0$. 

Types of random variable:

- If a sample space contains a finite number of possibilities or an unending sequence with as many elements as there are whole numbers, it is called a **discrete sample space**
- If a sample space contains an infinite number of possibilities equal to the number of points on a line segment, it is called a **continuous sample space**.

Probability mass function (p.m.f)

The set of ordered pairs $(x, f(x))$ is a **probability function**, **probability mass function**, or **probability distribution** of the discrete random variable X if, for each possible outcome x ,

1. $f(x) \geq 0$,
2. $\sum_x f(x) = 1$,
3. $P(X = x) = f(x)$.

The **cumulative distribution function** $F(x)$ of a discrete random variable X with probability distribution $f(x)$ is

$$F(x) = P(X \leq x) = \sum_{t \leq x} f(t), \quad \text{for } -\infty < x < \infty.$$

Discrete Probability distribution

Q: *A shipment of 20 similar laptop computers to a retail outlet contains 3 that are defective. If a school makes a random purchase of 2 of these computers, find the probability distribution for the number of defectives.*

Note :Also, you can discuss the CDF, $E(X)$, $Var(X)$, $E(aX+b)$, and $Var(aX+b)$ for the above question.

Example 5.2. A random variable X has the following probability distribution :

$x :$	0	1	2	3	4	5	6	7
$p(x) :$	0	k	$2k$	$2k$	$3k$	k^2	$2k^2$	$7k^2 + k$

(i) Find k , (ii) Evaluate $P(X < 6)$, $P(X \geq 6)$, and $P(0 < X < 5)$, (iii) If $P(X \leq c) > \frac{1}{2}$, find the minimum value of c , and (iv) Determine the distribution function of X .

[Madurai Univ. B.Sc., Oct. 1988]

Note :Also, you can discuss the $E(X)$, $Var(X)$, $E(aX+b)$, and $Var(aX+b)$ for the above question.

7. If $p(x) = \frac{x}{15}; x = 1, 2, 3, 4, 5$
 $= 0$, elsewhere

Find (i) $P\{X = 1 \text{ or } 2\}$, and (ii) $P\left\{\frac{1}{2} < X < \frac{5}{2} \mid X > 1\right\}$

[Allahabad Univ. B.Sc., April 1992]

Note :Also, you can discuss the CDF, $E(X)$, $\text{Var}(X)$, $E(aX+b)$, and $\text{Var}(aX+b)$ for the above question.

Practice problems

Q: Let W be a random variable giving the number of heads minus the number of tails in three tosses of a coin. List the elements of the sample space S for the three tosses of the coin and to each sample point assign a value w of W .

Note :Also, you can discuss the CDF, $E(X)$, $Var(X)$, $E(aX+b)$, and $Var(aX+b)$ for the above question.

Q: Determine the value c so that each of the following functions can serve as a probability distribution of the discrete random variable X :

$$f(x) = c \binom{2}{x} \binom{3}{3-x}, \text{ for } x = 0, 1, 2.$$

Q: The probability distribution of X , the number of imperfections per 10 meters of a synthetic fabric in continuous rolls of uniform width, is given by

x	0	1	2	3	4
$f(x)$	0.41	0.37	0.16	0.05	0.01

Construct the cumulative distribution function of X .

Practice problems

Q: A shipment of 7 television sets contains 2 defective sets. A hotel makes a random purchase of 3 of the sets. If x is the number of defective sets purchased by the hotel, find the probability distribution of X .

Q: Three cards are drawn in succession from a deck without replacement. Find the probability distribution for the number of spades.

Note :Also, you can discuss the CDF, $E(X)$, $Var(X)$, $E(aX+b)$, and $Var(aX+b)$ for the above question.

Continuous Probability distribution

- The probability distribution of a continuous random variable cannot be presented in tabular form, it can be stated as a formula. Such a formula would necessarily be a function of the numerical values of the continuous random variable X and as such will be represented by the functional notation $f(x)$. In dealing with continuous variables, $f(x)$ is usually called the probability density function, or simply the density function, of X .

The function $f(x)$ is a **probability density function** (pdf) for the continuous random variable X , defined over the set of real numbers, if

1. $f(x) \geq 0$, for all $x \in R$.
2. $\int_{-\infty}^{\infty} f(x) \, dx = 1$.
3. $P(a < X < b) = \int_a^b f(x) \, dx$.

The **cumulative distribution function** $F(x)$ of a continuous random variable X with density function $f(x)$ is

$$F(x) = P(X \leq x) = \int_{-\infty}^x f(t) \, dt, \quad \text{for } -\infty < x < \infty.$$

Example 5.4. A continuous random variable X has a p.d.f.

$$f(x) = 3x^2, \quad 0 \leq x \leq 1. \text{ Find } a \text{ and } b \text{ such that}$$

(i) $P\{X \leq a\} = P\{X > a\}$, and

(ii) $P\{X > b\} = 0.05$.

[Calicut Univ. B.Sc., Sept. 1988]

(iii) Determine the median value (Note: The value of 'a' obtained in part (i) represents the median)"

Note :Also, you can discuss the CDF, $E(X)$, $Var(X)$, $E(aX+b)$, and $Var(aX+b)$ for the above question.

Q: Suppose a certain type of small data processing firm is so specialized that some have difficulty making a profit in their first year of operation. The probability density function that characterizes the proportion Y that make a profit is given by

$$f(y) = \begin{cases} ky^4(1-y)^3, & 0 \leq y \leq 1, \\ 0, & \text{elsewhere.} \end{cases}$$

- (a) What is the value of k that renders the above a valid density function?
- (b) Find the probability that at most 50% of the firms make a profit in the first year.
- (c) Find the probability that at least 80% of the firms make a profit in the first year.

Note :Also, you can discuss the CDF, $E(X)$, $Var(X)$, $E(aX+b)$, and $Var(aX+b)$ for the above question.

Practice problems

Q) The diameter, say X , of an electric cable, is assumed to be a continuous random variable with p.d.f: $f(x) = 6x(1 - x)$, $0 \leq x \leq 1$

(i) Check that the above is a p.d.f

(ii) Obtain an expression for c.d.f of X ,

(iii) Compute $P\left(X \leq \frac{1}{2} \mid \frac{1}{3} \leq X \leq \frac{2}{3}\right)$, and

(iv) Determine the number k such that $P(X < k) = P(X > k)$

Q) Let X be a continuous random variable with p.d.f:

$$f(x) = \begin{cases} ax, & 0 \leq x \leq 1 \\ a, & 1 \leq x \leq 2 \\ -ax + 3a, & 2 \leq x \leq 3 \\ 0, & \text{else where} \end{cases}$$

Determine the constant a , and compute $P(X \leq 1.5)$

Verify that the following is a distribution function

.

$$F(x) = \begin{cases} 0, & x < -a \\ \frac{1}{2}\left(\frac{x}{a} + 1\right), & -a \leq x \leq a \\ 1, & x > a. \end{cases}$$

Practice problems

Q: The waiting time, in hours, between successive speeders spotted by a radar unit is a continuous random variable with cumulative distribution function:

$$F(x) = \begin{cases} 0, & x < 0, \\ 1 - e^{-8x}, & x \geq 0. \end{cases}$$

Find the probability of waiting less than 12 minutes between successive speeders

- (a) using the cumulative distribution function of X;
- (b) using the probability density function of X.

Note :Also, you can discuss the CDF, $E(X)$, $Var(X)$, $E(aX+b)$, and $Var(aX+b)$ for the above question.

Practice problems

Suppose that the error in the reaction temperature, in °C, for a controlled laboratory experiment is a continuous random variable X having the probability density function

$$f(x) = \begin{cases} \frac{x^2}{3}, & -1 < x < 2, \\ 0, & \text{elsewhere.} \end{cases}$$

- (a) Verify that $f(x)$ is a density function.
- (b) Find $P(0 < X \leq 1)$.
- (c) Find the Cumulative density function and use it to evaluate $P(0 < X \leq 1)$.

Note :Also, you can discuss the CDF, $E(X)$, $Var(X)$, $E(aX+b)$, and $Var(aX+b)$ for the above question.

Practice problems

Q: The proportion of people who respond to a certain mail-order solicitation is a continuous random variable X that has the density function

$$f(x) = \begin{cases} \frac{2(x+2)}{5}, & 0 < x < 1, \\ 0, & \text{elsewhere.} \end{cases}$$

- (a) Show that $P(0 < X < 1) = 1$.
- (b) Find the probability that more than $1/4$ but fewer than $1/2$ of the people contacted will respond to this type of solicitation.

Q: The shelf life, in days, for bottles of a certain prescribed medicine is a random variable having the density function

$$f(x) = \begin{cases} \frac{20,000}{(x+100)^3}, & x > 0, \\ 0, & \text{elsewhere.} \end{cases}$$

Find the probability that a bottle of this medicine will have a shelf life of

- (a) at least 200 days;
- (b) anywhere from 80 to 120 days.

Expected value of random variable

Let X be a random variable with probability distribution $f(x)$. The **mean**, or **expected value**, of X is

$$\mu = E(X) = \sum_x x f(x)$$

if X is discrete, and

$$\mu = E(X) = \int_{-\infty}^{\infty} x f(x) dx$$

if X is continuous.

Example 6.1. Let X be a random variable with the following probability distribution :

x	:	-3	6	9
$P_r(X=x)$	:	$1/6$	$1/2$	$1/3$

Find $E(X)$ and $E(X^2)$ and using the laws of expectation, evaluate $E(2X+1)^2$.

(Gauhati Univ. B.Sc., 1992)

Q: A salesperson for a medical device company has two appointments on a given day. At the first appointment, he believes that he has a 70% chance to make the deal, from which he can earn \$1000 commission if successful. On the other hand, he thinks he only has a 40% chance to make the deal at the second appointment, from which, if successful, he can make \$1500. What is his expected commission based on his own probability belief? Assume that the appointment results are independent of each other.

Practice problems

Example 6.2. (a) Find the expectation of the number on a die when thrown.
 (b) Two unbiased dice are thrown. Find the expected values of the sum of numbers of points on them.

Example 6.5. A coin is tossed until a head appears. What is the expectation of the number of tosses required ?
 [Delhi Univ. B.Sc., Oct. 1989]

Suppose that the number of cars X that pass through a car wash between 4:00 P.M. and 5:00 P.M. on any sunny Friday has the following probability distribution:

x	4	5	6	7	8	9
$P(X = x)$	$\frac{1}{12}$	$\frac{1}{12}$	$\frac{1}{4}$	$\frac{1}{4}$	$\frac{1}{6}$	$\frac{1}{6}$

Let $g(X) = 2X - 1$ represent the amount of money, in dollars, paid to the attendant by the manager. Find the attendant's expected earnings for this particular time period.

Practice problems

A lot containing 7 components is sampled by a quality inspector; the lot contains 4 good components and 3 defective components. A sample of 3 is taken by the inspector. Find the expected value of the number of good components in this sample.

(ii) Find Variance of number of good components in sample.

Expected value of function of random variable

Let X be a random variable with probability distribution $f(x)$. The expected value of the random variable $g(X)$ is

$$\mu_{g(X)} = E[g(X)] = \sum_x g(x)f(x)$$

if X is discrete, and

$$\mu_{g(X)} = E[g(X)] = \int_{-\infty}^{\infty} g(x)f(x) \, dx$$

if X is continuous.

Expected value of function of two random variable

Let X and Y be random variables with joint probability distribution $f(x, y)$. The mean, or expected value, of the random variable $g(X, Y)$ is

$$\mu_{g(X,Y)} = E[g(X, Y)] = \sum_x \sum_y g(x, y) f(x, y)$$

if X and Y are discrete, and

$$\mu_{g(X,Y)} = E[g(X, Y)] = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} g(x, y) f(x, y) \, dx \, dy$$

if X and Y are continuous.

Properties of expectation

- If a and b are constants, then

$$E(aX + b) = aE(X) + b.$$

- The expected value of the sum or difference of two or more functions of a random variable X is the sum or difference of the expected values of the functions. That is,

$$E[g(X) \pm h(X)] = E[g(X)] \pm E[h(X)].$$

- Let X and Y be two independent random variables. Then

$$E(XY) = E(X)E(Y).$$

- Let X and Y be two independent random variables. Then $\sigma_{XY} = 0$.

- If X and Y are random variables with joint probability distribution $f(x, y)$ and a , b , and c are constants, then

$$\sigma_{aX+bY+c}^2 = a^2\sigma_X^2 + b^2\sigma_Y^2 + 2ab\sigma_{XY}.$$

Practice problems

4.11 The density function of coded measurements of the pitch diameter of threads of a fitting is

$$f(x) = \begin{cases} \frac{4}{\pi(1+x^2)}, & 0 < x < 1, \\ 0, & \text{elsewhere.} \end{cases}$$

Find the expected value of X .

4.20 A continuous random variable X has the density function

$$f(x) = \begin{cases} e^{-x}, & x > 0, \\ 0, & \text{elsewhere.} \end{cases}$$

Find the expected value of $g(X) = e^{2X/3}$.

4.13 The density function of the continuous random variable X , the total number of hours, in units of 100 hours, that a family runs a vacuum cleaner over a period of one year, is given as

$$f(x) = \begin{cases} x, & 0 < x < 1, \\ 2 - x, & 1 \leq x < 2, \\ 0, & \text{elsewhere.} \end{cases}$$

Find the average number of hours per year that families run their vacuum cleaners.

Variance of random variable

Let X be a random variable with probability distribution $f(x)$ and mean μ . The variance of X is

$$\sigma^2 = E[(X - \mu)^2] = \sum_x (x - \mu)^2 f(x), \quad \text{if } X \text{ is discrete, and}$$
$$\sigma^2 = E[(X - \mu)^2] = \int_{-\infty}^{\infty} (x - \mu)^2 f(x) dx, \quad \text{if } X \text{ is continuous.}$$

The positive square root of the variance, σ , is called the **standard deviation** of X .

The variance of a random variable X is

$$\sigma^2 = E(X^2) - \mu^2.$$

Practice problems

4.34 Let X be a random variable with the following probability distribution:

x	-2	3	5
$f(x)$	0.3	0.2	0.5

Find the standard deviation of X .

4.50 For a laboratory assignment, if the equipment is working, the density function of the observed outcome X is

$$f(x) = \begin{cases} 2(1-x), & 0 < x < 1, \\ 0, & \text{otherwise.} \end{cases}$$

Find the variance and standard deviation of X .

Joint probability distribution

If X and Y are two discrete random variables, the probability distribution for their simultaneous occurrence can be represented by a function with values $f(x, y)$ for any pair of values (x, y) within the range of the random variables X and Y . It is customary to refer to this function as the **joint probability distribution** of X and Y .

Hence, in the discrete case,

$$f(x, y) = P(X = x, Y = y);$$

that is, the values $f(x, y)$ give the probability that outcomes x and y occur at the same time. For example, if an 18-wheeler is to have its tires serviced and X represents the number of miles these tires have been driven and Y represents the number of tires that need to be replaced, then $f(30000, 5)$ is the probability that the tires are used over 30,000 miles and the truck needs 5 new tires.

Discrete Joint Probability Distribution

The function $f(x, y)$ is a **joint probability distribution** or **probability mass function** of the discrete random variables X and Y if

1. $f(x, y) \geq 0$ for all (x, y) ,
2. $\sum_x \sum_y f(x, y) = 1$,
3. $P(X = x, Y = y) = f(x, y)$.

For any region A in the xy plane, $P[(X, Y) \in A] = \sum \sum_A f(x, y)$.

Continuous joint probability distribution

The function $f(x, y)$ is a **joint density function** of the continuous random variables X and Y if

1. $f(x, y) \geq 0$, for all (x, y) ,
2. $\int_{-\infty}^{\infty} \int_{-\infty}^{\infty} f(x, y) \, dx \, dy = 1$,
3. $P[(X, Y) \in A] = \int \int_A f(x, y) \, dx \, dy$, for any region A in the xy plane.

Marginal probability distribution

The **marginal distributions** of X alone and of Y alone are

$$g(x) = \sum_y f(x, y) \quad \text{and} \quad h(y) = \sum_x f(x, y)$$

for the discrete case, and

$$g(x) = \int_{-\infty}^{\infty} f(x, y) \, dy \quad \text{and} \quad h(y) = \int_{-\infty}^{\infty} f(x, y) \, dx$$

for the continuous case.

Conditional probability distribution

Let X and Y be two random variables, discrete or continuous. The **conditional distribution** of the random variable Y given that $X = x$ is

$$f(y|x) = \frac{f(x, y)}{g(x)}, \text{ provided } g(x) > 0.$$

Similarly, the conditional distribution of X given that $Y = y$ is

$$f(x|y) = \frac{f(x, y)}{h(y)}, \text{ provided } h(y) > 0.$$

1. Calculate the Conditional Mean:

- For a random variable X , the conditional mean given $Y = y$ is denoted as $E[X|Y = y]$.
- Similarly, for a random variable Y , the conditional mean given $X = x$ is denoted as $E[Y|X = x]$.

2. Calculate the Conditional Variance:

- The conditional variance of X given $Y = y$ is:

$$\text{Var}(X|Y = y) = E[(X - E[X|Y = y])^2|Y = y]$$

- The conditional variance of Y given $X = x$ is:

$$\text{Var}(Y|X = x) = E[(Y - E[Y|X = x])^2|X = x]$$

1. For Discrete Random Variables:

If X and Y are discrete random variables, then the conditional expectation is given by:

$$E[X \mid Y = y] = \sum_x x \cdot P(X = x \mid Y = y)$$

where $P(X = x \mid Y = y)$ is the conditional probability of $X = x$ given $Y = y$.

If X and Y are discrete random variables, the conditional expectation is given by:

$$E[Y \mid X = x] = \sum_y y \cdot P(Y = y \mid X = x)$$

where $P(Y = y \mid X = x)$ is the conditional probability of $Y = y$ given $X = x$.

2. For Continuous Random Variables:

If X and Y are continuous random variables, then the conditional expectation is given by:

$$E[X \mid Y = y] = \int_{-\infty}^{\infty} x \cdot f_{X|Y}(x \mid y) dx$$

where $f_{X|Y}(x \mid y)$ is the conditional probability density function of X given $Y = y$.

In both cases, the conditional expectation $E[X \mid Y = y]$ is a function of y .

If X and Y are continuous random variables, the conditional expectation is given by:

$$E[Y \mid X = x] = \int_{-\infty}^{\infty} y \cdot f_{Y|X}(y \mid x) dy$$

where $f_{Y|X}(y \mid x)$ is the conditional probability density function of Y given $X = x$.

In both cases, the conditional expectation $E[Y \mid X = x]$ is a function of x .

1. Conditional Variance $V[X \mid Y = y]$:

The conditional variance of X given $Y = y$ is defined as:

$$V[X \mid Y = y] = E[(X - E[X \mid Y = y])^2 \mid Y = y]$$

This can also be expressed as:

$$V[X \mid Y = y] = E[X^2 \mid Y = y] - (E[X \mid Y = y])^2$$

where:

- $E[X^2 \mid Y = y]$ is the conditional expectation of X^2 given $Y = y$.
- $E[X \mid Y = y]$ is the conditional expectation of X given $Y = y$.

2. Conditional Variance $V[Y \mid X = x]$:

The conditional variance of Y given $X = x$ is defined as:

$$V[Y \mid X = x] = E[(Y - E[Y \mid X = x])^2 \mid X = x]$$

This can also be expressed as:

$$V[Y \mid X = x] = E[Y^2 \mid X = x] - (E[Y \mid X = x])^2$$

where:

- $E[Y^2 \mid X = x]$ is the conditional expectation of Y^2 given $X = x$.
- $E[Y \mid X = x]$ is the conditional expectation of Y given $X = x$.

3. Find the Total Variance Using the Law of Total Variance:

- For the random variable X :

$$\text{Var}(X) = E[\text{Var}(X|Y)] + \text{Var}(E[X|Y])$$

- Here, $E[\text{Var}(X|Y)]$ is the expected value of the conditional variance of X given Y .
 - $\text{Var}(E[X|Y])$ is the variance of the conditional expectation of X given Y .
- For the random variable Y :

$$\text{Var}(Y) = E[\text{Var}(Y|X)] + \text{Var}(E[Y|X])$$

- Here, $E[\text{Var}(Y|X)]$ is the expected value of the conditional variance of Y given X .
- $\text{Var}(E[Y|X])$ is the variance of the conditional expectation of Y given X .

Consider the following joint probability distribution for two discrete random variables X and Y :

X	Y	$P(X = x, Y = y)$
0	0	0.1
0	1	0.2
1	0	0.3
1	1	0.4

1. Calculate the conditional probability $P(X = 0|Y = 0)$ and $P(X = 1|Y = 0)$.
2. Find the conditional means $E[X|Y = 0]$ and $E[X|Y = 1]$.
3. Compute the conditional variances $\text{Var}(X|Y = 0)$ and $\text{Var}(X|Y = 1)$.
4. Calculate the total variance of X using the law of total variance.
5. Similarly, compute the total variance of Y using the law of total variance.

1. Conditional Probabilities:

- For $Y = 0$:

$$P(X = 0|Y = 0) = \frac{P(X = 0, Y = 0)}{P(Y = 0)} = \frac{0.1}{0.1 + 0.3} = 0.25$$

$$P(X = 1|Y = 0) = \frac{P(X = 1, Y = 0)}{P(Y = 0)} = \frac{0.3}{0.4} = 0.75$$

- For $Y = 1$:

$$P(X = 0|Y = 1) = \frac{P(X = 0, Y = 1)}{P(Y = 1)} = \frac{0.2}{0.2 + 0.4} = 0.33$$

$$P(X = 1|Y = 1) = \frac{P(X = 1, Y = 1)}{P(Y = 1)} = \frac{0.4}{0.6} = 0.67$$

2. Conditional Means:

- For $Y = 0$:

$$E[X|Y = 0] = (0 \times 0.25) + (1 \times 0.75) = 0.75$$

- For $Y = 1$:

$$E[X|Y = 1] = (0 \times 0.33) + (1 \times 0.67) = 0.67$$

3. Conditional Variances:

- For $Y = 0$:

$$E[X^2|Y = 0] = (0^2 \times 0.25) + (1^2 \times 0.75) = 0.75$$

$$\text{Var}(X|Y = 0) = E[X^2|Y = 0] - (E[X|Y = 0])^2 = 0.75 - 0.75^2 = 0.1875$$

- For $Y = 1$:

$$E[X^2|Y = 1] = (0^2 \times 0.33) + (1^2 \times 0.67) = 0.67$$

$$\text{Var}(X|Y = 1) = E[X^2|Y = 1] - (E[X|Y = 1])^2 = 0.67 - 0.67^2 \approx 0.2223$$

4. **Total Variance of X :**

$$E[\text{Var}(X|Y)] = (0.4 \times 0.1875) + (0.6 \times 0.2223) = 0.2091$$

$$\text{Var}(E[X|Y]) = (0.4 \times 0.75^2) + (0.6 \times 0.67^2) - (0.4 \times 0.75 + 0.6 \times 0.67)^2 \approx 0.0045$$

$$\text{Var}(X) = E[\text{Var}(X|Y)] + \text{Var}(E[X|Y]) = 0.2091 + 0.0045 \approx 0.2136$$

5. **Total Variance of Y :** Follow a similar procedure to calculate the total variance of Y .

Consider the following joint probability density function (PDF) for two continuous random variables X and Y :

$$f_{X,Y}(x,y) = \begin{cases} 6xy, & \text{if } 0 \leq x \leq 1 \text{ and } 0 \leq y \leq 1 \\ 0, & \text{otherwise} \end{cases}$$

1. Find the marginal PDFs $f_X(x)$ and $f_Y(y)$.
2. Determine the conditional PDFs $f_{Y|X}(y|x)$ and $f_{X|Y}(x|y)$.
3. Compute the conditional means $E[Y|X = x]$ and $E[X|Y = y]$.
4. Calculate the conditional variances $\text{Var}(Y|X = x)$ and $\text{Var}(X|Y = y)$.
5. Find the total variance of X and Y using the law of total variance.

1. Marginal PDFs:

- Marginal PDF of X :

$$f_X(x) = \int_0^1 6xy \, dy = 3x, \quad 0 \leq x \leq 1$$

- Marginal PDF of Y :

$$f_Y(y) = \int_0^1 6xy \, dx = 3y, \quad 0 \leq y \leq 1$$

2. Conditional PDFs:

- Conditional PDF of Y given $X = x$:

$$f_{Y|X}(y|x) = \frac{f_{X,Y}(x,y)}{f_X(x)} = \frac{6xy}{3x} = 2y, \quad 0 \leq y \leq 1$$

- Conditional PDF of X given $Y = y$:

$$f_{X|Y}(x|y) = \frac{f_{X,Y}(\downarrow y)}{f_Y(y)} = \frac{6xy}{3y} = 2x, \quad 0 \leq x \leq 1$$

3. Conditional Means:

- $E[Y|X = x]$:

$$E[Y|X = x] = \int_0^1 y \cdot 2y \, dy = \int_0^1 2y^2 \, dy = \frac{2}{3}$$

- $E[X|Y = y]$:

$$E[X|Y = y] = \int_0^1 x \cdot 2x \, dx = \int_0^1 2x^2 \, dx = \frac{2}{3}$$

4. Conditional Variances:

- $\text{Var}(Y|X = x)$:

$$E[Y^2|X = x] = \int_0^1 y^2 \cdot 2y \, dy = \frac{2}{4} = 0.5$$

$$\text{Var}(Y|X = x) = E[Y^2|X = x] - (E[Y|X = x])^2 = 0.5 - \left(\frac{2}{3}\right)^2 \approx 0.0556$$

- $\text{Var}(X|Y = y)$:

$$E[X^2|Y = y] = \int_0^1 x^2 \cdot 2x \, dx = \frac{2}{4} = 0.5$$

$$\text{Var}(X|Y = y) = E[X^2|Y = y] - (E[X|Y = y])^2 = 0.5 - \left(\frac{2}{3}\right)^2 \approx 0.0556$$

5. Total Variances:

- $\text{Var}(Y)$:

$$\text{Var}(Y) = E[\text{Var}(Y|X)] + \text{Var}(E[Y|X]) = 0.0556 + \text{Var}\left(\frac{2}{3}\right) \approx 0.0556$$

- $\text{Var}(X)$:

$$\text{Var}(X) = E[\text{Var}(X|Y)] + \text{Var}(E[X|Y]) = 0.0556 + \text{Var}\left(\frac{2}{3}\right) \approx 0.0556$$

Statistically Independence

Let X and Y be two random variables, discrete or continuous, with joint probability distribution $f(x, y)$ and marginal distributions $g(x)$ and $h(y)$, respectively. The random variables X and Y are said to be **statistically independent** if and only if

$$f(x, y) = g(x)h(y)$$

for all (x, y) within their range.

Covariance of random variable

Let X and Y be random variables with joint probability distribution $f(x, y)$. The covariance of X and Y is

$$\sigma_{XY} = E[(X - \mu_X)(Y - \mu_Y)] = \sum_x \sum_y (x - \mu_X)(y - \mu_Y) f(x, y)$$

if X and Y are discrete, and

$$\sigma_{XY} = E[(X - \mu_X)(Y - \mu_Y)] = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} (x - \mu_X)(y - \mu_Y) f(x, y) \, dx \, dy$$

if X and Y are continuous.

The covariance of two random variables X and Y with means μ_X and μ_Y , respectively, is given by

$$\sigma_{XY} = E(XY) - \mu_X \mu_Y.$$

Example 6.32. Let X and Y be two random variables each taking three values $-1, 0$ and 1 , and having the joint probability distribution :

(i) Show that X and Y have different expectations.

$Y \backslash X$	-1	0	1	Total
-1	$\frac{0}{4}$	$\frac{1}{4}$	$\frac{1}{4}$	$\frac{2}{4}$
0	$\frac{2}{4}$	$\frac{2}{4}$	$\frac{2}{4}$	$\frac{6}{4}$
1	$\frac{0}{4}$	$\frac{1}{4}$	$\frac{1}{4}$	$\frac{2}{4}$
Total	$\frac{2}{4}$	$\frac{4}{4}$	$\frac{4}{4}$	$\frac{10}{4}$

(ii) Prove that X and Y are uncorrelated.

(iii) Find $\text{Var } X$ and $\text{Var } Y$.

(iv) Given that $Y = 0$, what is the conditional probability distribution of X ?

(v) Find $V(Y | X = -1)$.

Note: Part (ii) of **Example 6.32** is not relevant to Unit 1. You will be able to solve it after completing Unit 2

3.43 Let X denote the reaction time, in seconds, to a certain stimulus and Y denote the temperature ($^{\circ}\text{F}$) at which a certain reaction starts to take place. Suppose that two random variables X and Y have the joint density

$$f(x, y) = \begin{cases} 4xy, & 0 < x < 1, \ 0 < y < 1, \\ 0, & \text{elsewhere.} \end{cases}$$

Find

- (a) $P(0 \leq X \leq \frac{1}{2} \text{ and } \frac{1}{4} \leq Y \leq \frac{1}{2})$;
- (b) $P(X < Y)$.

Practice problems

Example 5.20. For the following bivariate probability distribution of X and Y , find

- (i) $P(X \leq 1, Y = 2)$, (ii) $P(X \leq 1)$, (iii) $P(Y = 3)$, (iv) $P(Y \leq 3)$ and (v) $P(X < 3, Y \leq 4)$

$X \backslash Y$	1	2	3	4	5	6
0	0	0	$\frac{1}{32}$	$\frac{2}{32}$	$\frac{2}{32}$	$\frac{3}{32}$
1	$\frac{1}{16}$	$\frac{1}{16}$	$\frac{1}{8}$	$\frac{1}{8}$	$\frac{1}{8}$	$\frac{1}{8}$
2	$\frac{1}{32}$	$\frac{1}{32}$	$\frac{1}{64}$	$\frac{1}{64}$	0	$\frac{2}{64}$

The fraction X of male runners and the fraction Y of female runners who compete in marathon races are described by the joint density function

$$f(x, y) = \begin{cases} 8xy, & 0 \leq y \leq x \leq 1, \\ 0, & \text{elsewhere.} \end{cases}$$

3.49 Let X denote the number of times a certain numerical control machine will malfunction: 1, 2, or 3 times on any given day. Let Y denote the number of times a technician is called on an emergency call. Their joint probability distribution is given as

$f(x, y)$		x		
		1	2	3
y	1	0.05	0.05	0.10
	3	0.05	0.10	0.35
	5	0.00	0.20	0.10

- Evaluate the marginal distribution of X .
- Evaluate the marginal distribution of Y .
- Find $P(Y = 3 \mid X = 2)$.

Find the covariance of X and Y .

Practice problems

Two ballpoint pens are selected at random from a box that contains 3 blue pens, 2 red pens, and 3 green pens. If X is the number of blue pens selected and Y is the number of red pens selected, find

- (a) the joint probability function $f(x, y)$,
- (b) $P[(X, Y) \in A]$, where A is the region $\{(x, y) | x + y \leq 1\}$.

3.38 If the joint probability distribution of X and Y is given by

$$f(x, y) = \frac{x + y}{30}, \quad \text{for } x = 0, 1, 2, 3; \ y = 0, 1, 2,$$

- find
- (a) $P(X \leq 2, Y = 1)$;
 - (b) $P(X > 2, Y \leq 1)$;
 - (c) $P(X > Y)$;
 - (d) $P(X + Y = 4)$.

Practice problems

Q) The diameter ,say X , of an electric cable, is assumed to be a continuous random variable with p.d.f: $f(x) = 6x(1 - x), 0 \leq x \leq 1$

- (i) Check that the above is a p.d.f
- (ii) Obtain an expression for c.d.f of X ,
- (iii) Compute $P\left(X \leq \frac{1}{2} \mid \frac{1}{3} \leq X \leq \frac{2}{3}\right)$, and
- (iv) Determine the number k such that $P(X < k) = P(X > k)$

Q) Let X be a continuous random variable with p.d.f:

$$f(x) = \begin{cases} ax, & 0 \leq x \leq 1 \\ a, & 1 \leq x \leq 2 \\ -ax + 3a, & 2 \leq x \leq 3 \\ 0, & \text{else where} \end{cases}$$

Determine the constant a , and compute $P(X \leq 1.5)$

Verify that the following is a distribution function

$$F(x) = \begin{cases} 0, & x < -a \\ \frac{1}{2} \left(\frac{x}{a} + 1 \right), & -a \leq x \leq a \\ 1, & x > a. \end{cases}$$

Note :Also, you can discuss the CDF, $E(X)$, $Var(X)$, $E(aX+b)$, and $Var(aX+b)$ for the above question.

Practice problems

Example 5.30. If X and Y are two random variables having joint density function

$$f(x, y) = \frac{1}{8} (6 - x - y) ; 0 < x < 2, 2 < y < 4 \\ = 0, \text{ otherwise}$$

Find (i) $P(X < 1 \cap Y < 3)$, (ii) $P(X + Y < 3)$ and (iii) $P(X < 1 | Y < 3)$
(Madras Univ. B.Sc., Nov. 1986)

Practice problems

3.47 The amount of kerosene, in thousands of liters, in a tank at the beginning of any day is a random amount Y from which a random amount X is sold during that day. Suppose that the tank is not resupplied during the day so that $x \leq y$, and assume that the joint density function of these variables is

$$f(x, y) = \begin{cases} 2, & 0 < x \leq y < 1, \\ 0, & \text{elsewhere.} \end{cases}$$

- (a) Determine if X and Y are independent.
(b) Find $P(1/4 < X < 1/2 \mid Y = 3/4)$.

(iii) $Var(X)$ and $Var(Y)$.

(iv) Co-variance between X and Y .

Practice problems

Q: Two ballpoint pens are selected at random from a box that contains 3 blue pens, 2 red pens, and 3 green pens. Two refills for a ballpoint pen are selected at random from a certain box, and the following is the joint probability distribution:

$f(x, y)$		x			$h(y)$
		0	1	2	
y	0	$\frac{3}{28}$	$\frac{9}{28}$	$\frac{3}{28}$	$\frac{15}{28}$
	1	$\frac{3}{14}$	$\frac{3}{14}$	0	$\frac{3}{7}$
	2	$\frac{1}{28}$	0	0	$\frac{1}{28}$
$g(x)$		$\frac{5}{14}$	$\frac{15}{28}$	$\frac{3}{28}$	1

Find the covariance of X and Y .

Chebyshev's theorem

In probability theory, Chebyshev inequality guarantees, that for a wide class of probability distribution (Binomial, Poisson, normal,...), NO more than a certain fraction of values can be more than a certain distance from the mean.

Specifically, no more than $1/(k^2)$ of the distribution values can be more than k standard deviation away from the mean (or equivalently, at least $1 - 1/(k^2)$ of the distribution values are within k standard deviation of the mean)

Chebyshev's theorem

: (Chebyshev's Theorem) The probability that any random variable X will assume a value within k standard deviations of the mean is at least $1 - 1/k^2$. That is,

$$P(\mu - k\sigma < X < \mu + k\sigma) \geq 1 - \frac{1}{k^2}.$$

OR

$$P(|X - \mu| < k\sigma) \geq 1 - \frac{1}{k^2}$$

Also for $P(|X - \mu| \geq k\sigma) \leq \frac{1}{k^2}$

Remark:-

By Chebyshev's Theorem

- $\frac{3}{4}$ (i.e. 75%) or more of the data of any distribution lies within two standard deviations of the mean i.e. in the interval $(\mu \pm 2\sigma)$
- $\frac{8}{9}$ (or 88.9%) or more of the data of any distribution lies within three standard deviations of the mean i.e. in the interval $(\mu \pm 3\sigma)$
- $\frac{15}{16}$ (or 93.75%) or more of the data of any distribution lies within four standard deviations of the mean i.e. in the interval $(\mu \pm 4\sigma)$

.....

$(1 - \frac{1}{k^2})$ or more of the data of any distribution lies within k standard deviations of the mean i.e. in the interval $(\mu \pm k\sigma)$

4.77 A random variable X has a mean $\mu = 10$ and a variance $\sigma^2 = 4$. Using Chebyshev's theorem, find

- (a) $P(|X - 10| \geq 3)$;
- (b) $P(|X - 10| < 3)$;
- (c) $P(5 < X < 15)$;
- (d) the value of the constant c such that

$$P(|X - 10| \geq c) \leq 0.04.$$

4.78 Compute $P(\mu - 2\sigma < X < \mu + 2\sigma)$, where X has the density function

$$f(x) = \begin{cases} 6x(1-x), & 0 < x < 1, \\ 0, & \text{elsewhere,} \end{cases}$$

and compare with the result given in Chebyshev's theorem.

Practice problems

Example 4.27: A random variable X has a mean $\mu = 8$, a variance $\sigma^2 = 9$, and an unknown probability distribution. Find

- (a) $P(-4 < X < 20)$,
- (b) $P(|X - 8| \geq 6)$.

Example : The average score in Mathematics exam is given as 84 with standard deviation 4. Find the range of marks between which at least 75% of the students scored.

4.75 An electrical firm manufactures a 100-watt light bulb, which, according to specifications written on the package, has a mean life of 900 hours with a standard deviation of 50 hours. At most, what percentage of the bulbs fail to last even 700 hours? Assume that the distribution is symmetric about the mean.